



HNDIT4242-Computer Services Management



Lecture 09

Availability Management – Resource Scheduling & Asset Management



Lesson Objectives

- Explain **resource scheduling** and its role in IT service management.
- Understand **IT asset management** and its importance in maintaining infrastructure.
- Learn how IT supports **library administration and digital asset management**.
- Explore techniques for **team organization and IT service optimization**.
- Apply best practices for **efficient IT resource utilization**.



1. Resource Scheduling & IT Asset Management

- ***1.1 What is Resource Scheduling in IT?***
 - **Definition:** The process of allocating IT resources (hardware, software, staff, cloud services) efficiently.
 - **Importance:** Ensures optimal **resource utilization**, prevents downtime, and improves service delivery.
 - **Use Cases:**
 - ✓ Scheduling **server maintenance** to minimize disruptions.
 - ✓ Allocating **bandwidth and computing power** for cloud services.
 - ✓ Managing **IT support staff shifts** for 24/7 availability.



1.2 Best Practices for IT Resource Scheduling

- 1. Automation:** Use scheduling software (e.g., **Microsoft System Center, ServiceNow**) to automate repetitive tasks.
- 2. Prioritization:** Assign resources based on urgency and impact (e.g., **critical server maintenance over minor UI updates**).
- 3. Load Balancing:** Distribute workloads across multiple servers to prevent overload.
- 4. Monitoring:** Track resource usage in real time using tools like **Zabbix, SolarWinds, or AWS CloudWatch**.



2. IT Asset Management (ITAM)

2.1 What is IT Asset Management?

- **Definition:** The process of tracking, managing, and maintaining IT assets throughout their lifecycle.
- **Types of IT Assets:**
 - ✓ **Hardware** (Servers, Laptops, Printers, Routers)
 - ✓ **Software** (Operating Systems, Applications, Cloud Services)
 - ✓ **Networks** (Firewalls, VPNs, Storage)
 - ✓ **Human Resources** (IT Staff, Technicians)

2.2 IT Asset Lifecycle Management

IT assets go through different stages:

Stage	Description
Procurement	Acquiring new IT resources
Deployment	Installing and configuring assets
Utilization	Active use in production environments
Maintenance	Regular updates, patches, and repairs
Retirement	Decommissioning outdated assets



2.3 Best Practices in IT Asset Management

1. **Centralized Asset Inventory:** Maintain a **single database** of all IT assets.
2. **Regular Audits:** Conduct periodic **hardware and software audits** to track performance and compliance.
3. **Software License Management:** Ensure legal compliance by tracking **software licenses** (e.g., **Microsoft Volume Licensing, Adobe CC**).
4. **Security & Risk Management:** Protect IT assets from **cyber threats, hardware failures, and unauthorized access**.
5. **Cloud Asset Management:** Monitor and optimize cloud resource usage in platforms like **AWS, Azure, and Google Cloud**.



3. Library Administration & Digital Asset Management

3.1 Role of IT in Library Administration

- Traditional **libraries** are transitioning to **digital libraries** with IT-driven solutions.
- IT enables:
 - ✓ **Online library catalogs** (e.g., Koha, Evergreen ILS).
 - ✓ **E-book and digital content management** (e.g., Google Books, Open Library).
 - ✓ **Automated circulation and book tracking systems.**
 - ✓ **Secure authentication for user access.**



3.2 Digital Asset Management (DAM)

- **Definition:** A system to store, organize, and retrieve digital content (PDFs, images, videos, software).
- **IT Role:**
 - ✓ **Metadata tagging** for better searchability.
 - ✓ **Cloud-based access** for scalability.
 - ✓ **Data backup and disaster recovery** for content protection.



4. Organizing IT Support Teams for Efficiency

4.1 Importance of IT Team Organization

- IT support teams ensure **smooth operations** in businesses, universities, and government organizations.
- Key roles in an IT team:
 - ✓ **IT Administrators** (Manage servers, databases, and user accounts).
 - ✓ **Help Desk Technicians** (Handle user support and troubleshooting).
 - ✓ **Cybersecurity Experts** (Protect against threats and enforce security policies).
 - ✓ **Network Engineers** (Manage connectivity, bandwidth, and security protocols).



4.2 Strategies for Effective Team Organization

- 1. Shift Scheduling:** Ensure 24/7 IT support availability.
- 2. Role Specialization:** Assign specific responsibilities (e.g., database admin vs. security analyst).
- 3. Collaboration Tools:** Use platforms like **Microsoft Teams, Slack, or Jira** for better communication.
- 4. Incident Management Frameworks:** Implement ITSM methodologies (**ITIL, COBIT**) to handle issues efficiently.



Summary

- **Resource scheduling** ensures IT resources are optimally utilized, reducing downtime.
- **IT asset management** tracks and maintains hardware, software, and network resources.
- **Digital libraries** enhance accessibility and organization of knowledge resources.
- **Well-organized IT teams** improve efficiency, reduce response time, and ensure business continuity.



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